

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250280578

Kind Code

A1

Publication Date

September 04, 2025

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CONTACT STRUCTURES FOR SEMICONDUCTOR DEVICES

Abstract

The present disclosure describes semiconductor devices and methods for forming the same. A method for forming a semiconductor device includes forming a source/drain structure and forming a gate structure. The method also includes performing a cleaning process on the source/drain structure and the gate structure. The method also includes disposing a portion of a byproduct of the cleaning process on a top surface of the gate structure and etching the portion of the byproduct so a remaining portion of the byproduct is formed on the top surface of the gate structure. The method further includes forming a gate contact structure, including depositing a metal material on the remaining portion of the byproduct to form a compound containing the metal material and the remaining portion of the byproduct. The method also includes forming a barrier layer between the compound and the top surface of the gate structure.

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Appl. No.: 19/201200

Filed: May 07, 2025

Related U.S. Application Data

parent US division 17716331 20220408 parent-grant-document US 12317554 child US 19201200
us-provisional-application US 63173169 20210409

Publication Classification

Int. Cl.: H10D62/10 (20250101); H10D30/67 (20250101); H10D84/01 (20250101); H10D84/03 (20250101)

U.S. Cl.:

CPC H10D62/118 (20250101); H10D30/6713 (20250101); H10D30/6735 (20250101); H10D30/6757 (20250101); H10D84/0128 (20250101); H10D84/013 (20250101); H10D84/038 (20250101);

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a divisional of U.S. Non-provisional patent application Ser. No. 17/716,331, titled “Contact Structures for Semiconductor Devices,” which was filed on Apr. 8, 2022, which claims the benefit of U.S. Provisional Patent Appl. No. 63/173,169, titled “A Novel Contact Structure” and filed on Apr. 9, 2021, both of which are incorporated herein by reference in their entireties.

BACKGROUND

[0002] With advances in semiconductor technology, there has been increasing demand for higher storage capacity, faster processing systems, higher performance, and lower costs. To meet these demands, the semiconductor industry continues to scale down the dimensions of semiconductor devices and three-dimensional transistors, such as gate-all-around field effect transistors (GAAFETs) and fin field effect transistors (finFETs), have been introduced.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0003] Aspects of this disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the common practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

[0004] FIG. 1 is a flow diagram of a method for fabricating contact structures having fluorine-metal compounds, in accordance with some embodiments.

[0005] FIGS. 2A-2D, 3A, 3B, and 4-7 illustrate various cross-sectional views of semiconductor devices at various stages of their fabrication process, in accordance with some embodiments.

[0006] FIGS. 8-13 illustrate various cross-sectional views of semiconductor devices incorporating contact structures having fluorine-metal compounds at various stages of their fabrication process, in accordance with some embodiments.

[0007] Illustrative embodiments will now be described with reference to the accompanying drawings. In the drawings, like reference numerals generally indicate identical, functionally similar, and/or structurally similar elements.

DETAILED DESCRIPTION

[0008] The following disclosure provides different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features are disposed between the first and second features, so the first and second

features are not in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

[0009] Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

[0010] The acronym “FET,” as used herein, refers to a field effect transistor. An example of a FET is a metal oxide semiconductor field effect transistor (MOSFET). MOSFETs can be, for example, (i) planar structures built in and on the planar surface of a substrate, such as a semiconductor wafer, or (ii) built with vertical structures.

[0011] The term “nominal” as used herein refers to a desired, or target, value of a characteristic or parameter for a component or a process operation, set during the design phase of a product or a process, together with a range of values above and/or below the desired value. The range of values is typically due to slight variations in manufacturing processes or tolerances.

[0012] The terms “about” and “substantially” as used herein indicate the value of a given quantity that can vary based on a particular technology node associated with the subject semiconductor device. In some embodiments, based on the particular technology node, the terms “about” and “substantially” can indicate a value of a given quantity that varies within, for example, 5% of the value (e.g., $\pm 1\%$, $\pm 2\%$, $\pm 3\%$, $\pm 4\%$, $\pm 5\%$ of the value), 10% of the value, 20% of the value, etc.

[0013] The present disclosure provides example field effective transistor (FET) devices (e.g., gate-all-around (GAA) FETs, fin-type FET (finFETs), horizontal or vertical GAA finFETs, or planar FETs) in a semiconductor device and/or in an integrated circuit (IC) and example methods for fabricating the same.

[0014] The fins of finFETs may be patterned by any suitable method. For example, the fins may be patterned using one or more photolithography processes, including double-patterning or multi-patterning processes. Generally, double-patterning or multi-patterning processes combine photolithography and self-aligned processes, allowing patterns to be created that have, for example, pitches smaller than what is otherwise obtainable using a single, direct photolithography process. For example, in some embodiments, a sacrificial layer is formed over a substrate and patterned using a photolithography process. Spacers are formed alongside the patterned sacrificial layer using a self-aligned process. The sacrificial layer is then removed, and the remaining spacers may then be used to pattern the fins.

[0015] GAAFETs and finFETs increase device density and improve device performance.

GAAFETs and finFETs include a pair of source/drain regions formed on opposite sides of a channel region. The channel region is covered by a gate structure including gate dielectric layers, work function layers, and a metal gate electrode. The gate structure can be formed using a replacement gate process where a polysilicon gate is disposed on the channel region and replaced by a metal gate structure. Source/drain contacts and gate contacts are respectively formed on the source/drain regions and gate structures to provide electrical bias for controlling the transistor devices. Contact resistances for the source/drain contact and gate contacts are factors that can impact device performance. For example, a lower contact resistance is desired for achieving lower power consumption and faster operating speed. However, impedance match between the transistor devices and external circuitry may also play an important role in achieving nominal device performance, and it may be desirable to incorporate gate contact structures having similar impedances as the impedance of external circuitry connected to the gate structure. As the semiconductor industry continues to scale down the dimensions of semiconductor devices, circuit

complexity has increased at all device levels. For example, beyond the 5 nm technology node or the 3 nm technology node, decreased gate length can increase the difficulty of achieving nominal gate contact resistance.

[0016] Various embodiments in the present disclosure describe methods for forming gate contact structures having nominal resistances. Specifically, a cleaning process performed on the gate structures to remove one or more oxide materials can dispose fluorine-containing material, such as ammonium hydrogen fluoride (NH₄F), on top surfaces of the gate electrode. One or more treatment processes can remove portions of the fluorine-containing material so a nominal amount of fluorine-containing material remains on the top surface of the gate electrode. Metal material, such as titanium, is subsequently deposited on the top surface of the gate electrode and reacts with the fluorine-containing material to form a fluorine-containing metal compound. Conductive vias, such as tungsten, can be deposited on the fluorine-containing metal compound. In some embodiments, the amount of fluorine-containing metal compound formed between the gate electrode and the conductive via can be determined by the amount of fluorine-containing material after the treatment process and the amount of subsequently-deposited metal material. A greater amount of fluorine-containing metal compound can provide a gate contact structure having a greater resistance. The gate contact structure described in the present application provides various benefits, such as adjustable gate contact resistance by varying the treatment and deposition processing conditions. The gate contact resistance can be increased or decreased to match the impedance of subsequently formed external circuitry, which in turn improves device performance and reliability. The embodiments described herein use GAAFETs as examples and can be applied to other semiconductor structures, such as finFETs and planar FETs. In addition, the embodiments described herein can be used in various technology nodes, such as 14 nm, 7 nm, 5 nm, 3 nm, 2 nm, and lower technology nodes.

[0017] FIG. 1 is a flow diagram of a method **100** for fabricating a semiconductor device incorporating resistance-adjustable gate contacts, according to some embodiments. For illustrative purposes, the operations illustrated in FIG. 1 will be described with reference to the example fabrication process of fabricating a semiconductor device **200** as illustrated in FIGS. 2A-2D, 3A, 3B, and 4-13. Operations can be performed in a different order or not performed depending on specific applications. It should be noted that method **100** may not produce a complete semiconductor device. Accordingly, it is understood that additional processes can be provided before, during, and after method **100**, and that some other processes may only be briefly described herein.

[0018] Referring to FIG. 1, in operation **105**, semiconductor layers are formed on fin structures of a substrate, according to some embodiments. For example, fin structure **108** with fin base portion **108A** and fin top portion **108B** can be formed on substrate **106** as described with reference to semiconductor device **200** illustrated in FIGS. 2A-2C. FIG. 2B is a cross-sectional view of the structure in FIG. 2A as viewed from the A-A line. FIG. 2C is a cross-sectional view of the structure in FIG. 2A as viewed from the B-B line. The formation of fin structure **108** can include the formation of fin base portion **108A** and fin top portion **108B** on substrate **106**. FIGS. 2A-2C illustrate semiconductor layers formed in a wire configuration (e.g., cross-sectional area having a substantially square shape). Alternatively, semiconductor layers of semiconductor device **200** can also be formed in a sheet configuration (e.g., cross-sectional area having a substantially rectangular shape), as illustrated in FIG. 2D.

[0019] Substrate **106** can be a semiconductor material, such as silicon. In some embodiments, substrate **106** includes a crystalline silicon substrate (e.g., wafer). In some embodiments, substrate **106** includes (i) an elementary semiconductor, such as germanium; (ii) a compound semiconductor including silicon carbide, gallium arsenide, gallium phosphide, indium phosphide, indium arsenide, and/or indium antimonide; (iii) an alloy semiconductor including silicon germanium carbide, silicon germanium, gallium arsenic phosphide, gallium indium phosphide, gallium indium

arsenide, gallium indium arsenic phosphide, aluminum indium arsenide, and/or aluminum gallium arsenide; or (iv) a combination thereof. Further, substrate **106** can be doped depending on design requirements (e.g., p-type substrate or n-type substrate). In some embodiments, substrate **106** can be doped with p-type dopants (e.g., boron, indium, aluminum, or gallium) or n-type dopants (e.g., phosphorus or arsenic).

[0020] Fin structure **108** extends along an x-axis. Fin structure **108** can be a part of a substrate and include a fin base portion **108A** and a fin top portion **108B** disposed on fin base portion **108A**.

[0021] Fin base portion **108A** can include material similar to substrate **106**. Fin base portion **108A** can be formed from a photolithographic patterning and an etching of substrate **106**. Fin top portion **108B** can include a stack of semiconductor layers. Each semiconductor layer can be subsequently processed to form a channel region underlying subsequently-formed gate structures of the finFETs.

[0022] Fin top portion **108B** includes a first group of semiconductor layers **122** and a second group of semiconductor layers **124** stacked in an alternating configuration. Each of semiconductor layers **122** and **124** can be epitaxially grown on its underlying layer and can include semiconductor materials different from each other. In some embodiments, semiconductor layers **122** and **124** can include semiconductor materials similar to or different from substrate **106**. In some embodiments, semiconductor layers **122** and **124** can include semiconductor materials with oxidation rates and/or etch selectivities different from each other. In some embodiments, each of semiconductor layers **122** can be formed of silicon and each of semiconductor layers **124** can be formed of silicon germanium. In some embodiments, semiconductor layers **122** can be formed of silicon germanium and semiconductor layers **124** can be formed of silicon. Semiconductor layers **122** and/or semiconductor layers **124** can be undoped or can be in-situ doped during their epitaxial growth process using (i) p-type dopants, such as boron, indium, and gallium; and/or (ii) n-type dopants, such as phosphorus and arsenic. For p-type in-situ doping, p-type doping precursors, such as diborane, boron trifluoride, and any other p-type doping precursor, can be used. For n-type in-situ doping, n-type doping precursors, such as phosphine and arsine, can be used. Though four layers for each of semiconductor layers **122** and semiconductor layers **124** are shown in FIGS. 2A-2C, semiconductor device **200** can have any suitable number of semiconductor layers **122** and semiconductor layers **124**.

[0023] Forming fin base portion **108A** and fin top portion **108B** can also include etching the aforementioned stack of materials through patterned hard mask layers **134** and **136** formed on the stack of materials. In some embodiments, hard mask layer **134** can be a thin film including silicon oxide formed using, for example, a thermal oxidation process. In some embodiments, hard mask layer **136** can be formed of silicon nitride using, for example, low-pressure chemical vapor deposition (LPCVD) or plasma-enhanced chemical vapor deposition (PECVD). The etching of the stack of materials can include a dry etch, a wet etch process, or a combination thereof. Hard mask layers **134** and **136** can be removed after fin structures **108** are formed.

[0024] Fin top portions **108B** can be formed using stacks of semiconductor layers **122** and **124** in a wire configuration, as shown in FIG. 2C. For example, semiconductor layers **122** and **124** in FIG. 2C have a substantially square-shaped cross-sectional area. In some embodiments, fin top portions **108B** can be formed using stacks of semiconductor layers **122** and **124** in a sheet configuration, as shown in FIG. 2D. For example, semiconductor layers **122** and **124** in FIG. 2C have a substantially rectangular-shaped cross-sectional area. FIGS. 3A, 3B, and 4-8 illustrate forming semiconductor devices with semiconductor layers in a wire configuration that subsequently form nanosheet structures. The method illustrated in the present disclosure also applies to semiconductor devices with semiconductor layers in the sheet configuration that subsequently form nanosheet structures.

[0025] Referring to FIG. 1, in operation **110**, sacrificial gate structures are formed on the substrate and the semiconductor layers are etched, according to some embodiments. Referring to FIGS. 3A and 3B, STI regions **138** with first and second protective liners **138A** and **138B** and insulating layer **138C** can be formed on substrate **106**. FIG. 3B is a cross-sectional view of semiconductor device

200 in FIG. 3A as viewed from the C-C line. In some embodiments, substrate **106** can include fin bottom portion **108A** and are collectively referred to as substrate **106** for simplicity. In some embodiments, hard mask layer **136** remains on the top surfaces of hard mask layer **134** after the formation of STI regions **138**. In some embodiments, hard mask layer **136** is removed prior to the formation of STI regions **138**. Forming STI regions **138** can include (i) depositing a layer of nitride material (not shown) for first protective liners **138A** on the structure of FIG. 2A, (ii) depositing a layer of oxide material (not shown) for second protective liners **138B** on the layer of nitride material, (iii) depositing a layer of insulating material for insulating layers **138C** on the layer of oxide material, (iv) annealing the layer of insulating material for insulating layer **138C**, (v) chemical mechanical polishing (CMP) the layers of nitride and oxide materials and the annealed layer of insulating material, and (vi) etching back the polished structure to form the structure of FIG. 3A. The layers of nitride and oxide materials can be deposited using a suitable process for depositing oxide and nitride materials, such as atomic layer deposition (ALD) and chemical vapor deposition (CVD). These layers of oxide and nitride materials can prevent oxidation of the sidewalls of fin top portion **108B** during the deposition and annealing of the insulating material for insulating layer **138C**. In some embodiments, the layer of insulating material for insulating layer **138C** can include silicon oxide, silicon nitride, silicon oxynitride, fluoride-doped silicate glass (FSG), or a low-k dielectric material. In some embodiments, the layer of insulating material can be deposited using a CVD process, a high-density-plasma (HDP) CVD process, using silane and oxygen as reacting precursors. In some embodiments, the layer of insulating material can be formed using a sub-atmospheric CVD (SACVD) process or high aspect-ratio process (HARP), where process gases can include tetraethoxysilane (TEOS) and/or ozone.

[0026] Polysilicon gate structures **112** are formed on STI regions **138**, as shown in FIGS. 3A and 3B. Polysilicon gate structures **112** are sacrificial gate structures and can be replaced in a gate replacement process to form metal gate structures. In some embodiments, the formation of polysilicon gate structures **112** can include blanket depositing a layer of polysilicon material and etching the layer of polysilicon material through a patterned hard mask layer **116** formed on the layer of polysilicon material. In some embodiments, the layer of polysilicon material can be undoped and hard mask layer **116** can include an oxide layer and/or a nitride layer. The oxide layer can be formed using a thermal oxidation process and the nitride layer can be formed by LPCVD or PECVD. Hard mask layer **116** can protect polysilicon gate structures **112** from subsequent processing steps (e.g., during formation of spacers **114** and/or source/drain regions). The blanket deposition of the layer of polysilicon material can include CVD, physical vapor deposition (PVD), ALD, or any other suitable deposition process. In some embodiments, etching of the deposited layer of polysilicon material can include a dry etch, a wet etch, or a combination thereof. Spacers **114** can be formed on sidewalls of polysilicon gate structures **112**. Forming spacers **114** can include blanket depositing a layer of an insulating material (e.g., an oxide, a nitride, and/or silicon carbon oxynitride material) followed by photolithography and an etching process (e.g., reactive ion etching or any other suitable dry etching process using a chlorine-or fluorine-based etchant).

[0027] Fin top portions can be etched after polysilicon gate structures **112** are formed. The etch process can remove portions of semiconductor layers **122** and **124** that are exposed between adjacent polysilicon gate structures **112**. In some embodiments, the etching process can be a cyclic etching process for removing materials that form semiconductor layers **122** and **124**, such as etching processes for removing silicon and silicon germanium materials. For example, the etch process can include a wet etch process using, for example, diluted hydrofluoric acid for etching silicon germanium and tetramethylammonium hydroxide (TMAH) for etching silicon material. In some embodiments, one or more etching processes can be used. During the etching process, polysilicon gate structures **112** can be protected by spacers **114** and hard mask layer **116**.

[0028] Referring to FIG. 1, in operation **115**, inner spacer structures are formed between the semiconductor layers, according to some embodiments. Referring to FIG. 4, portions of

semiconductor layers **124** can be etched back to form recessed regions and dielectric material can be deposited in the recessed regions to form inner spacers **127**. For example, semiconductor device **200** shown in FIG. **4** can include n-type metal-oxide-semiconductor (NMOS) devices and portions of semiconductor layers **124** are etched back. In some embodiments, substrate **106** can include fin bottom portion **108A** and are collectively referred to as substrate **106** for simplicity.

[0029] Semiconductor device **200** illustrated in FIG. **4** can include semiconductor layers **124** and substrate **106** formed using silicon germanium and semiconductor layers **122** formed using silicon. In some embodiments, substrate **106** can be an intrinsic material or doped with suitable dopants. For example, substrate **106** can have a non-uniform dopant concentration. Semiconductor device **200** can also include p-type metal-oxide-semiconductor (PMOS) devices. PMOS device configurations are not shown in FIG. **4** for simplicity. For the PMOS device configurations, semiconductor layers **124** can be processed to be used as the channel regions. Semiconductor layers **122** can be etched back using suitable etching processes and inner spacers **127** can be formed between adjacent semiconductor layers **124** using similar deposition and etching processes described below with respect to the etch back of semiconductor layers **124** and the formation of inner spacers **127**.

[0030] Semiconductor layers **124** can be etched back by a dry etching process, a wet etching process, or a combination thereof. The etch back process of semiconductor layers **124** can be configured to form non-planar outer surfaces of semiconductor layers **122** and **124**. For example, the etching process can include alternating cycles of etching and purging processes. The etching process in each cycle can include using a gas mixture having hydrogen fluoride, nitrogen trifluoride, a fluorine-based gas, and/or a chlorine-based gas. As shown in enlarged view **401** of FIG. **4**, semiconductor layers **122** can have curved convex outer surfaces **122t** and semiconductor layers **124** can have curved concave outer surfaces **124t**. In some embodiments, subsequently-formed inner spacers **127** can also have outer surfaces **127t** that substantially contour outer surface **124t** of semiconductor layers **124**.

[0031] Referring to FIG. **1**, in operation **120**, source/drain structures are formed on the semiconductor layers and the inner spacer structures, according to some embodiments. Referring to FIG. **5**, source/drain structures **502** can be formed on outer surfaces of semiconductor layers **122** and inner spacers **127**. Source/drain structures **502** are also in contact with top surfaces of substrate **106**. In some embodiments, source/drain structure **502** can be formed by a selective growth process where a semiconductor material is grown on selective surfaces. For example, source/drain structure **502** can be formed by epitaxially growing a crystalline material using top surfaces of substrate **106** as seed layers.

[0032] Source/drain structures **502** can be formed of silicon, silicon germanium, silicon phosphide, any suitable semiconductor material, and combinations thereof. In some embodiments, source/drain structures **502** can be doped with suitable dopants, such as boron and phosphorus. Source/drain structures **502** can be formed using suitable deposition or growth methods, such as (i) CVD, including but not limited to, LPCVD, atomic layer CVD (ALCVD), ultrahigh vacuum CVD (UHVCVD), reduced pressure CVD (RPCVD), and any other suitable CVD; (ii) molecular beam epitaxy (MBE) processes; (iii) any suitable epitaxial process; and (iv) a combination thereof. In some embodiments, source/drain structure **502** can be grown by an epitaxial deposition/partial etch process, which repeats the epitaxial deposition/partial etch process at least once. Such repeated deposition/partial etch process is also called a cyclic deposition-etch (CDE) process. In some embodiments, a plasma deposition process using species, such as germane, dichlorosilane, and hydrochloride, can be used to deposit source/drain structure **502** formed of silicon germanium. A width *W* of source/drain structure **502** can be between about 10 nm and about 80 nm, between about 15 nm and about 75 nm, between about 20 nm and about 60 nm, or any suitable dimension. In some embodiments, a height *H* of source/drain structure **502** measured from top surface **502t** of source/drain structure **502** and bottom surface **502b** can be between about 20 nm and about 140

nm, between about 30 nm and about 120 nm, between about 40 nm and about 100 nm, or any suitable dimension.

[0033] Referring to FIG. 1, in operation 125, etching processes are performed and nanostructures are released (e.g., exposed), according to some embodiments. Referring to FIG. 6, semiconductor layers 124 are removed, exposing portions of semiconductor layer 122 formed between opposite inner spacers 127. The exposed semiconductor layers can be referred to as nanostructures (e.g., nanowires or nanosheets). In some embodiments, semiconductor layers 122 are removed (not illustrated in FIG. 6) and semiconductor layers 124 form nanostructures.

[0034] Prior to the release of the nanostructures, an interlayer dielectric (ILD) layer 618 can be deposited between spacers 114 and polysilicon gate structures 112 are removed, according to some embodiments. ILD layer 618 can be disposed on source/drain structure 502 of the source/drain regions and between spacers 114. ILD layer 618 can include a dielectric material deposited using a deposition method suitable for flowable dielectric materials (e.g., flowable silicon oxide, flowable silicon nitride, flowable silicon oxynitride, flowable silicon carbide, or flowable silicon oxycarbide). For example, the flowable silicon oxide can be deposited using flowable CVD (FCVD). In some embodiments, the dielectric material is silicon oxide. Other materials and formation methods for ILD layer 618 are within the scope and spirit of this disclosure.

[0035] The formation of ILD layer 618 can be followed by removing polysilicon gate structures 112 and semiconductor layers 124 using a dry etching process (e.g., reaction ion etching) or a wet etching process and exposing portions of semiconductor layers 122. The exposed semiconductor layers 122 can be referred to as nanostructures (e.g., nanowires or nanosheets). Depending on the type of devices being formed, semiconductor layers 122 can be removed, exposing portions of semiconductor layers 124, which can also be referred to as nanostructures. In some embodiments, the gas etchants used in the dry plasma etching process can include hydrogen and radicals, such as chlorine, fluorine, bromine, and a combination thereof. For example, the gas etchants can include hydrogen bromide, hydrogen chloride, or any suitable gas etchant. In some embodiments, wet chemical etching can be used. Etchants for the wet chemical etching process can include ozone mixed with one or more of hydrogen fluoride, hydrogen chloride, hydrogen peroxide, or any suitable chemical etchant. In some embodiments, a dry etch followed by a wet etch process can be used.

[0036] Referring to FIG. 1, in operation 130, gate dielectric layers, work function layers, and gate electrodes are deposited on the nanostructures, according to some embodiments. Referring to FIG. 7, a gate stack 710 including gate dielectric layers 712, work function layers 714, and gate electrode 716 are formed on semiconductor layers 122 and on substrate 106. Gate dielectric layers 712 can be formed on the semiconductor layers. In some embodiments, gate dielectric layers 712 can be wrapped around exposed nanostructure-shaped semiconductor layers 122. In some embodiments, semiconductor layers 122 can be nanosheets or nanowires. Forming gate dielectric layers 712 can include a blanket deposition process of a suitable gate dielectric material layer. In some embodiments, gate dielectric layers 712 can be formed of a high-k dielectric material (e.g., dielectric material having dielectric constant greater than about 3.9). For example, gate dielectric layers 712 can be formed of hafnium oxide. In some embodiments, one or more gate dielectric layers can be formed. Work function layers 714 are formed on gate dielectric layers 712. In some embodiments, each work function layer 714 can include one or more work function metal layers and be formed using the same or different material and/or thickness. In some embodiments, work function layers can include titanium nitride and/or titanium aluminum alloy. Gate dielectric layers 712 and gate work function layers 714 can each wrap around nanostructure-shaped semiconductor layers 122. Depending on the spaces between adjacent semiconductor layers 122, semiconductor layers 122 can be wrapped around by gate dielectric layer 712 and work function layers 714, filling the spaces between adjacent semiconductor layers 122. In some embodiments, subsequently-formed gate electrode material can also be formed in the spaces between adjacent semiconductor

layers **122**, as illustrated in enlarged view **750** and described below.

[0037] Gate electrodes **716** can be formed on the work function layers, according to some embodiments. Layers of conductive material for gate electrodes **716** are formed on work function layers **714**. As shown in enlarged view **750**, if separations between adjacent semiconductor layers **122** are sufficient to accommodate the thickness of the gate electrode material, gate electrodes **716** can be formed between adjacent semiconductor layers **122** and on work function layers **714** so the spaces between adjacent semiconductor layers **122** are filled. Gate electrodes **716** that are between adjacent semiconductor layers **122** and gate electrodes **716** that are formed between spacers **114** are electrically coupled to each other. The layer of conductive material for gate electrodes **716** can include suitable conductive materials, such as titanium, silver, aluminum, tungsten, copper, ruthenium, molybdenum, tungsten nitride, cobalt, nickel, titanium carbide, titanium aluminum carbide, manganese, zirconium, metal alloys, and combinations thereof. Gate electrodes **716** can be formed by ALD, PVD, CVD, or any other suitable deposition process. The deposition of gate electrodes **716** can continue until openings between opposite spacers **114** are filled with gate electrodes **716**. A chemical mechanical polishing process can remove excessive gate electrodes **716** so top surfaces of gate electrodes **716** and ILD layer **618** are substantially coplanar. In some embodiments, other structures can be formed, such as blocking layers. One or more blocking layers (not shown in FIG. 7) can be formed prior to depositing gate electrodes **716** to prevent diffusion and oxidation of gate electrodes **716**.

[0038] Enlarged view **750** illustrates portions of the gate structure formed between adjacent semiconductor layers **122**. As shown in enlarged view **750**, inner spacers **127** can have height $H_{sub.1}$ as measured between opposite surfaces from adjacent nanostructures, such as semiconductor layers **122**. In some embodiments, height $H_{sub.1}$ can be between about 3 nm and about 10 nm, between about 5 nm and about 7 nm, or any suitable height.

[0039] Referring to FIG. 1, in operation **135**, openings are formed to expose the source/drain structures and gate electrodes and one or more cleaning processes are performed, according to some embodiments. Referring to FIG. 8, etching processes are performed to form openings **804** and **806** to expose underlying source/drain structures **502** and gate electrodes **716**, respectively. One or more cleaning processes can be used to remove an oxide from exposed source/drain structures **502** and gate electrodes **716**. After the oxide is removed, byproducts **820** of the one or more cleaning processes are formed at top surfaces of source/drain structures **502** and gate electrodes **716**.

[0040] In some embodiments, dielectric layers **816** and **818** are deposited on ILD layer **618** and gate stack **710** prior to the formation of openings. In some embodiments, dielectric layers **816** and **818** can be formed using different materials and dielectric layer **816** can be an etch stop layer. In some embodiments, dielectric layer **816** can be formed using silicon oxynitride and dielectric layer **818** can be formed using silicon oxide. Dielectric layers **816** and **818** can be deposited using any suitable deposition method, such as CVD, PECVD, FCVD, LPCVD, ALD, and any combinations thereof.

[0041] In some embodiments, openings **804** and **806** can be formed by patterning and anisotropic etching processes that etches portions of dielectric layers **816** and **818** as well as ILD **618**. The etching processes can be a dry etching process for removing the respective dielectric material of the aforementioned layers. In some embodiments, reactive ion etch (RIE) etching processes or wet chemical etching processes can be used. In some embodiments, the chemical etchants can include ammonium hydroxide, hydrogen peroxide, hydrofluoric or alkaline-based etchant, sulfur hexafluoride, any suitable chemical etchant, and combinations thereof. The etching processes continues until top surfaces of source/drain structures **502** and gate electrode **716** are exposed. In some embodiments, openings **806** extend into gate electrode **716**. For example, a bottom surface of openings **806** is below a top surface of gate electrode **716**, as shown in FIG. 8.

[0042] The etching processes to form openings **804** and **806** may also oxidize the exposed top surfaces of source/drain structures **502** and gate electrode **716**. Therefore, a cleaning process can be

performed to remove the oxide. In some embodiments, the cleaning process can be performed using a plasma incorporating fluorinated chemical species. For example, the plasma process used for removing oxidization can use partially or fully fluorinated hydrocarbon, sulfur hexafluoride, and combinations thereof. In some embodiments, ammonium can also be mixed with the fluorinated chemical species and used during the cleaning process. Although the cleaning process can remove oxidization from the top surfaces of source/drain structures **502** and gate electrode **716**, it may also deposit fluoride-containing byproducts **820** on the top surfaces as a result of the chemical reactions between the fluorinated chemical species and the oxide material. In some embodiments, byproducts **820** can be a fluorine-containing material, such as ammonium hydrogen fluoride. In some embodiments, byproducts **820** can also contain one or more metal elements in addition to fluorine. In some embodiments, byproducts **820** within opening **806** can have a thickness $t_{sub.1}$ between about 30 Å and about 60 Å, between about 35 Å and about 55 Å, between about 40 Å and about 50 Å, or any suitable thickness. In some embodiments, byproducts **820** formed in openings **806** can have a greater thickness than byproducts **820** formed in openings **804**.

[0043] Referring to FIG. 1, in operation **140**, one or more treatment processes are performed to adjust a fluorine deposit level on the gate electrodes, according to some embodiments. Referring to FIG. 9, treatment processes **902** can be performed to remove byproducts **820** from openings **804** and **806**. As byproduct **820** can be formed using a fluorine-containing material, such as ammonium hydrogen fluoride, the fluorine deposit level can be adjusted by adjusting the amount of byproduct **820** remaining on the top surface of gate electrode **716**. For example, thickness $t_{sub.1}$ of byproduct **820** can be reduced to thickness $t_{sub.2}$ after treatment process **902**. In some embodiments, thickness $t_{sub.2}$ can be between about 20 Å and about 35 Å. For example, thickness $t_{sub.2}$ can be between about 22 Å and about 33 Å, between about 25 Å and about 30 Å, or any suitable thickness. In some embodiments, a ratio of $t_{sub.2}$ over $t_{sub.1}$ can be between about 30% and about 70%. In some embodiments, a greater thickness $t_{sub.2}$ can provide the benefit of having a variable (e.g., a greater) gate contact resistance, which in turn can be matched to an impedance of subsequently-formed contacts or interconnects. In some embodiments, openings **804** formed over source/drain structures **502** can have lower aspect ratios (e.g., a ratio of height over width of an opening) than openings **806** formed over gate electrodes **716**. The lower aspect ratio allows treatment process **902** to remove byproduct **820** at a greater removal rate from openings **804** than from openings **806**. Therefore, in some embodiments, byproduct **820** can be completely removed from openings **804** and partially removed from openings **806**. As byproduct **820** is a fluorine-containing material, adjusting the fluorine deposit level on gate electrode **716** can be achieved by controllably and partially removing byproduct **820** processes from openings **806**.

[0044] Byproduct **820** can be removed using any suitable process, such as a multi-cycle degas process performed at an elevated temperature. In some embodiments, the multi-cycle degas process is performed at a temperature between about 300° C. and about 500° C. For example, the multi-cycle degas process can be performed between about 380° C. and about 420° C. In some embodiments, the multi-cycle degas process can include an active plasma etching process performed in a reactive ion etching chamber. For example, the multi-cycle degas process can be argon plasma etching in the presence of a plasma-enhancing carrier gas, such as carbon dioxide, or an inert gas, such as helium, xenon, neon, and krypton. In some embodiments, the degas process can be performed in a physical vapor deposition (PVD) chamber, an RF sputtering chamber, a magnetron sputtering chamber, an ionized metal plasma (IMP) chamber, a self-ionized plasma (SIP) sputtering chamber, and the like. In some embodiments, the multi-cycle degas process can be performed in one or more cycles. For example, each cycle can include a plasma-activated etching process performed for a pre-determined amount of time, and the cycle can be repeated two or more times until a nominal amount of byproduct **820** remains on the top surface of gate electrode **716**. In some embodiments, each cycle can be performed for between about 40 s and about 100 s, between

about 50 s and about 90 s, between about 60 s and about 80 s, or any suitable amount of time. In some embodiments, the cycle can be performed between 2 and 5 times, such as 3 and 4 times. [0045] Referring to FIG. 1, in operation 145, fluorine-metal compounds and barrier layers are formed on the gate electrodes, according to some embodiments. Referring to FIG. 10, fluorine-metal compound 1002 and barrier layer 1004 are formed above gate electrodes 716 and between portions dielectric layers 816 and 818. In some embodiments, fluorine-metal compound 1002 and barrier layer 1004 extend into gate electrode 716. Gate contacts 1006 include fluorine-metal compound 1002 and barrier layer 1004.

[0046] Fluorine-metal compound 1002 can be formed by directly depositing a metal in opening 806 and on byproduct 820 (illustrated in FIG. 9 but not illustrated in FIG. 10). The metal deposition process can be a plasma-activated deposition method that activates byproduct 820 remaining in openings 806 so the deposited metal and byproduct 820 form a fluorine-metal compound. For example, the metal deposition process can be a plasma-activated deposition process for depositing titanium. In some embodiments, other suitable metal materials can be deposited. For example, a metal material such as tantalum, aluminum, copper, tungsten, or any suitable metal material can be deposited. In some embodiments, byproduct 820 can be formed using ammonium hydrogen fluoride and titanium can be deposited to form fluorine-metal compound 1002 containing ammonium hydrogen fluoride and titanium. In some embodiments, fluorine-metal compound 1002 can have a thickness $t_{sub.3}$ between about 5 nm and about 25 nm, between about 7 nm and about 22 nm, between about 10 nm and about 20 nm, or any other suitable thicknesses. In some embodiments, thickness $t_{sub.3}$ can be adjusted to provide a nominal contact resistance for gate contacts 1006. For example, a greater thickness $t_{sub.3}$ can provide a greater contact resistance and a lower thickness $t_{sub.3}$ can provide a lower contact resistance. In some embodiments, an optional barrier layer 1004 can be formed between fluorine-metal compound 1002 and dielectric layers 816 and 818. In some embodiments, barrier layer 1004 can be formed using a direct treatment process or a deposition process. In some embodiments, barrier layer 1004 can be formed using titanium nitride.

[0047] Referring to FIG. 1, in operation 150, source/drain contacts are formed on the source/drain regions, according to some embodiments. Referring to FIG. 11, source/drain contacts 1102 are formed in openings 804 and in contact with source/drain structures 502. Source/drain contacts 1102 and gate contacts 1006 are formed to provide electrical connections to the source/drain regions and the gate electrodes, respectively. Specifically, source/drain contacts 1102 and gate contacts 1006 can be used to transmit electrical signals between source/drain regions and gate electrodes and external terminals (not shown in FIG. 11). Source/drain contacts 1102 can be formed by depositing a conductive material in openings 804. The deposition process can include depositing a metal layer within openings 804 and performing an anneal process to induce silicidation of the deposited metal layer. The conductive materials for forming source/drain contacts 1102 can include cobalt. In some embodiments, source/drain contacts 1102 can be formed using titanium, aluminum, silver, tungsten, copper, ruthenium, zirconium, nickel, titanium nitride, tungsten nitride, metal alloys, and combinations thereof. The deposition process can include ALD, PVD, CVD, any suitable deposition process, and combinations thereof.

[0048] A planarization process can planarize the top surfaces of dielectric layer 818, source/drain contacts 1102, and gate contacts 1006 so the top surfaces are substantially coplanar. In some embodiments, source/drain contacts 1102 can extend into source/drain structures 502. Silicide regions (not shown in FIG. 11) can be formed between source/drain contacts 1102 and source/drain structures 502 to reduce contact resistance. In some embodiments, the silicide regions can include ruthenium silicide, nickel silicide, cobalt silicide, tungsten silicide, tantalum silicide, platinum silicide, erbium silicide, palladium silicide, any suitable silicide material, and combinations thereof.

[0049] Referring to FIG. 1, in operation 155, dielectric layers are deposited and interconnect structures are formed in the dielectric layers, according to some embodiments. Referring to FIG.

12, dielectric layer **1202** is deposited on the top surface of dielectric layer **818**, source/drain contacts **1102**, and gate contacts **1006**. Conductive interconnects, such as vias **1204** and **1206**, are formed on source/drain contacts **1102** and gate contacts **1006**, respectively. Dielectric layer **1202** can be an ILD layer and be formed using similar material as dielectric layers **816** and **818**. For example, dielectric layer **1202** can be formed using a low-k dielectric material (e.g., a dielectric material having dielectric constant less than about **3.9**).

[0050] Vias **1204** and **1206** can be conductive structures having semi-spherical or arcuate-shaped anchors **1204A** and **1206A** respectively embedded in the underlying source/drain contacts **1102** and gate contacts **1006**. In some embodiments, the purpose of having anchors is twofold: (i) provide a larger contact area between vias **1204** and **1206** and the underlying source/drain contacts **1102** and gate contacts **1006**, and (ii) prevent damage to the interconnects during a planarization process. For example, vias **1204** and **1206** can be “pulled-out” during a planarization process of the top surfaces of dielectric layer **1202** and vias **1204** and **1206**.

[0051] Vias **1204** and **1206** can be formed by performing a patterning and etching process to form openings through dielectric layer **1202** to expose the underlying contact structures and performing one or more isotropic etching process to extend the openings in the underlying contact structures. For example, the isotropic etching processes can be wet chemical etching processes that have substantially similar etching rates in the vertical direction (e.g., z direction) and in the horizontal direction (e.g., x direction) so the extended openings extend under a bottom surface of dielectric layer **1202**. For example, a width of anchor **1204A** embedded in source/drain contact **1102** can be greater than a width of via **1204** formed in dielectric layer **1202**. Similarly, a width of anchor **1206A** embedded in gate contact **1006** can be greater than a width of via **1206** formed in dielectric layer **1202**. FIG. **12** illustrates anchor **1206A** of via **1206** embedded in fluorine-metal compound **1002** and separated from, or not in contact with, barrier layer **1004**. Although not illustrated in FIG. **12**, the isotropic etching processes may also etch barrier layer **1004** so anchor **1206A** may extend into barrier layer **1004**. Vias **1204** and **1206** can be formed using any suitable conductive material. For example, vias **1204** and **1206** can be formed using tungsten. In some embodiments, vias **1204** and **1206** can be formed using titanium, aluminum, silver, cobalt, copper, ruthenium, zirconium, nickel, titanium nitride, tungsten nitride, metal alloys, and combinations thereof.

[0052] Vias **1204** and **1206** can be formed in different steps and/or using different processes. For example, via **1204** can be formed prior to the formation of via **1206** using a selective growth method and via **1206** can be formed by depositing conductive material using a deposited adhesive or seed layer. In some embodiments, source/drain contacts **1102** are formed using cobalt and openings are formed both on source/drain contacts **1102** and gate contacts **1006**. A tungsten material can be selectively grown on source/drain contacts **1102** using the cobalt material of source/drain contacts **1102** as a seed layer. The tungsten material may have minimal growth on the fluorine-metal compound material of gate contacts **1006**. In some embodiments, an adhesive layer **1208** can be deposited on gate contact **1006** and sidewalls of dielectric layers **1202**. Adhesive layer **1208** is used for facilitating tungsten growth so tungsten material is deposited over and electrically coupled to gate contacts **1006**. In some embodiments, adhesive layer **1208** can also be a seed layer from which tungsten is grown. In some embodiments, adhesive layer **1208** can be formed of boron or silicon and deposited using a conformal film deposition process, such as ALD, CVD, PECVD, any suitable film deposition process, and combinations thereof.

[0053] FIG. **12** illustrates via **1206** formed on adhesive layer **1208**. Alternatively, a metal conductive layer can be deposited on gate contact **1006** and via **1206** can be formed using a selective growth method using the metal conductive layer as a seed layer. FIG. **13** illustrates via **1206** formed on conductive structure **1302** using a selective growth method. During the cobalt deposition process for forming source/drain contacts **1102**, a portion of cobalt material can also be deposited on gate contact **1006** to form a conductive structure **1302**. The formation of vias **1204** and **1206** can be conducted simultaneously by selectively growing tungsten using source/drain

contacts **1102** and conductive structure **1302** as seed layers.

[0054] Back-end-of-line (BEOL) interconnect structures (not illustrated in FIGS. **12** and **13**) can be formed on dielectric layer **1202**, interconnects **1204** and **1206**, and other suitable structures. In some embodiments, the BEOL interconnects can be a network of electrical connections that include vias extending vertically (e.g., along the z-axis) and wires extending laterally (e.g., along the x-axis). Interconnect structures can provide electrical connections to interconnects **1204** and **1206**. In some embodiments, suitable passive and active semiconductor devices can be formed in dielectric layers **816**, **818**, and **1202** and are not illustrated for simplicity.

[0055] Various embodiments in the present disclosure describe methods for forming gate contact structures having nominal resistances. Specifically, a cleaning process performed on the gate structures can dispose fluorine-containing material, such as ammonium hydrogen fluoride (NH₄HF), on top surfaces of the gate electrode. One or more treatment processes can remove portions of the fluorine-containing material so a nominal amount of fluorine-containing material remains on the top surface of the gate electrode. Metal material, such as titanium, is subsequently deposited on the top surface of the gate electrode and reacts with the fluorine-containing material to form a fluorine-containing metal compound. A conductive via, such as tungsten, can be deposited on the fluorine-containing metal compound to form the bulk of the gate contact structure. In some embodiments, the amount of fluorine-containing metal compound formed between the gate electrode and the conductive via can be determined by the amount of fluorine-containing material after the treatment process and the amount of subsequently-deposited metal material. A greater amount of fluorine-containing metal compound can provide a gate contact structure having a greater resistance. The gate contact structure described in the present application provide various benefits, such as adjustable gate contact resistance by varying the treatment and deposition processing conditions. The gate contact resistance can be increased or decreased to match the impedance of subsequently-formed external circuitry, which in turn improves device performance and reliability.

[0056] In some embodiments, a method for forming a semiconductor device includes forming a source/drain structure and forming a gate structure. The method also includes performing a cleaning process on the source/drain structure and the gate structure. The method also includes disposing a portion of a byproduct of the cleaning process on a top surface of the gate structure and etching the portion of the byproduct so a remaining portion of the byproduct is formed on the top surface of the gate structure. The method further includes forming a gate contact structure, including depositing a metal material on the remaining portion of the byproduct to form a compound containing the metal material and the remaining portion of the byproduct. The method also includes forming a barrier layer between the compound and the top surface of the gate structure.

[0057] In some embodiments, a method includes forming nanostructures on a substrate and forming a source/drain structure in contact with the nanostructures. The method also includes forming a gate structure on the nanostructures and performing a cleaning process on the source/drain structure and the gate structure. The method also includes disposing a first portion of a byproduct of the cleaning process on a top surface of the source/drain structure and a second portion of the byproduct on a top surface of the gate structure. The method also includes removing the first portion of the byproduct from the top surface of the source/drain structure and etching the second portion of the byproduct so a remaining second portion of the byproduct is formed on the top surface of the gate structure. The method further includes depositing a metal material on the remaining second portion of the byproduct to form a compound, the compound including the metal material and the remaining second portion of the byproduct. The method further includes forming a barrier layer between the compound and the top surface of the gate structure.

[0058] In some embodiments, a semiconductor structure includes a gate structure on nanostructures and a gate contact structure on the gate structure. The gate contact structure includes a barrier layer

and a fluorine-metal compound formed on the barrier layer. The semiconductor structure also includes an interconnect formed on and electrically coupled to the fluorine-metal compound. [0059] The foregoing disclosure outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A semiconductor structure, comprising: a gate structure on a plurality of nanostructures; a gate contact structure on the gate structure, wherein the gate contact structure comprises: a barrier layer; and a fluorine-metal compound formed on the barrier layer; and an interconnect formed on and electrically coupled to the fluorine-metal compound.
2. The semiconductor structure of claim 1, wherein the interconnect comprises tungsten.
3. The semiconductor structure of claim 1, wherein the fluorine-metal compound comprises titanium and ammonium hydrogen fluoride.
4. The semiconductor structure of claim 1, wherein the interconnect comprises a semi-spherical or arcuate-shaped anchor formed in the fluorine-metal compound.
5. The semiconductor structure of claim 1, wherein the barrier layer comprises titanium nitride.
6. The semiconductor structure of claim 1, wherein the fluorine-metal compound and the barrier layer are surrounded by a dielectric layer.
7. The semiconductor structure of claim 1, wherein the interconnect comprises cobalt.
8. A structure, comprising: a plurality of nanostructures formed on a substrate; a source/drain structure in contact with the plurality of nanostructures; a gate structure formed on the plurality of nanostructures; a gate contact comprising a fluorine metal compound; a conductive layer formed on the gate contact; and a first conductive via disposed on the conductive layer.
9. The structure of claim 8, wherein the conductive layer comprises cobalt.
10. The structure of claim 8, wherein the first conductive via protrudes into the conductive layer.
11. The structure of claim 8, wherein the first conductive via comprises an adhesive liner layer.
12. The structure of claim 11, wherein the adhesive liner layer is in contact with the conductive layer.
13. The structure of claim 8, further comprising a source/drain contact that extends into the source/drain structure.
14. The structure of claim 13, further comprising a second conductive via formed on the source/drain contact.
15. The structure of claim 14, wherein the second conductive via comprises a semi-spherical or arcuate-shaped anchor formed in the source/drain contact.
16. The structure of claim 8, wherein the first conductive via comprises a semi-spherical or arcuate-shaped anchor that extends into the conductive layer.
17. A structure, comprising: a plurality of nanostructured layers; a source drain region in contact with the plurality of nanostructured layers; a gate structure formed on the plurality of nanostructured layers; and a gate contact structure that extends into the gate structure, wherein the gate contact structure comprises a fluorine-metal layer.
18. The structure of claim 17, further comprising a conductive via that extends into the gate contact structure.
19. The structure of claim 18, wherein the conductive via comprises a bow-shaped anchor that

extends into the gate contact structure.

20. The structure of claim 18, wherein the conductive via comprises an adhesive layer.
